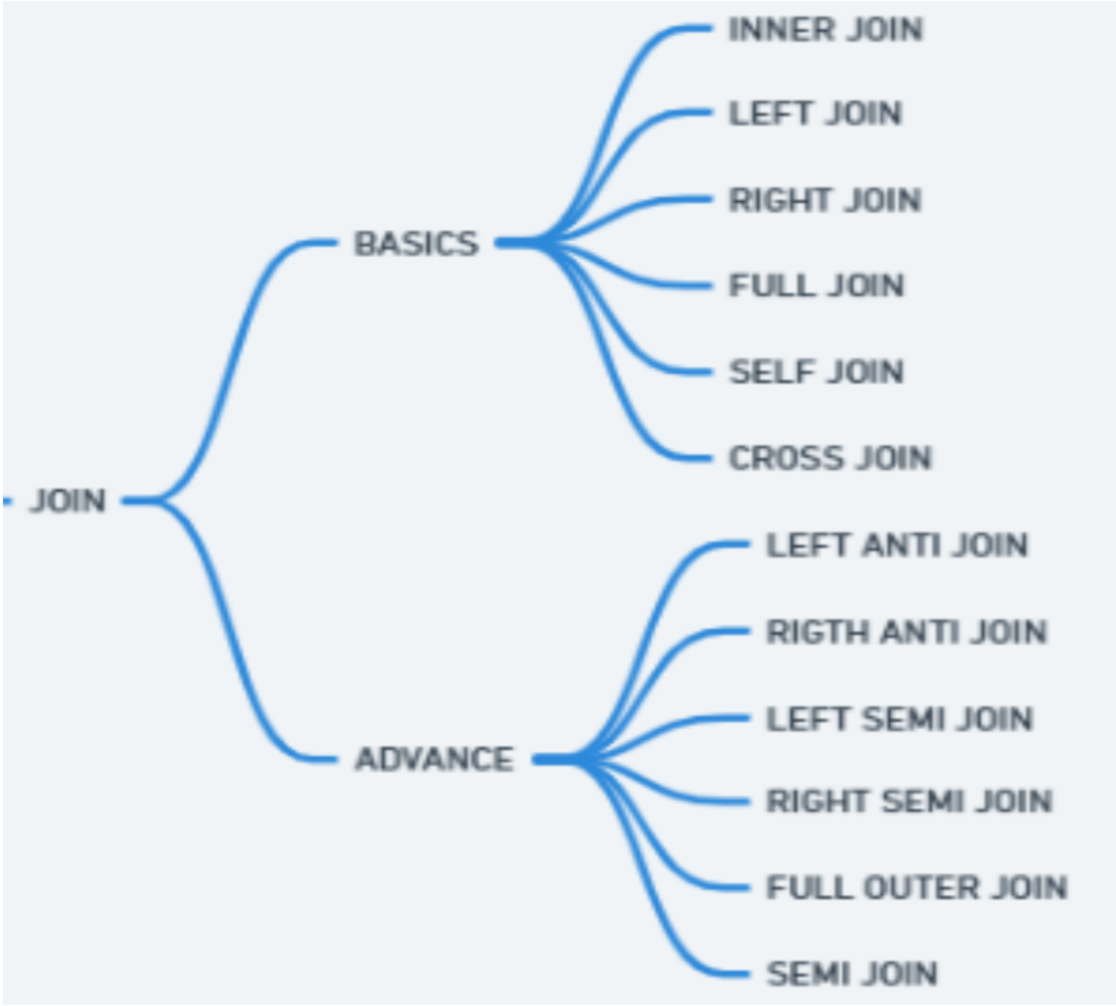


TABLE 1:-

EMPLOYEES

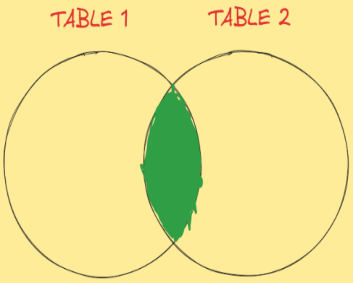
emp_id	name	dept_id
1	Ram	101
2	Shyam	102
3	Mohan	NULL
4	Sita	104

TABLE 2:-

DEPARTMENTS

dept_id	dept_name
101	HR
102	IT
103	Finance

1. INNER JOIN

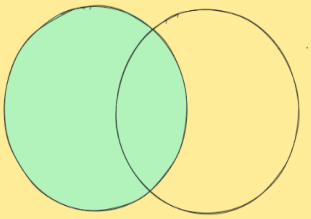


-- INNER JOIN JO DONO TABLES ME SE COMMON VALUE KO PRESENT KRTA HAI

-- SYNTEX :-
```
SELECT e.emp_id, e.name, d.dept_name
FROM Employees e
INNER JOIN Departments d
ON e.dept_id = d.dept_id;
```

emp_id	name	dept_name
1	Ram	HR
2	Shyam	IT

2. LEFT JOIN

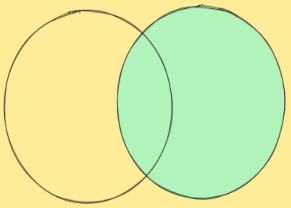


-- LEFT JOIN ME TABLE 1 VALI SARI VALUE AAYEGI OR TABLE 2 ME JO COMMON VALUE OUTPUT ME AAYEGI

-- SYNTEX :-
```
SELECT e.emp_id, e.name, d.dept_name
FROM Employees e
LEFT JOIN Departments d
ON e.dept_id = d.dept_id;
```

emp_id	name	dept_name
1	Ram	HR
2	Shyam	IT
3	Mohan	NULL
4	Sita	NULL

3. RIGHT JOIN

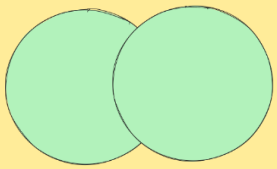


-- RIGHT JOIN ME TABLE 2 VALI SARI VALUE AAYEGI OR TABLE 1 ME JO COMMON VALUE OUTPUT ME AAYEGI

-- SYNTEX :-
```
SELECT e.emp_id, e.name, d.dept_name
FROM Employees e
RIGHT JOIN Departments d
ON e.dept_id = d.dept_id;
```

emp_id	name	dept_name
1	Ram	HR
2	Shyam	IT
NULL	NULL	Finance

4. FULL JOIN



-- FULL JOIN ME TABLE 1 VALI SARI VALUE AAYEGI OR TABLE 2 KI BHI SARI VALUE OUTPUT ME AAYEGI

-- SYNTEX :-
```
SELECT e.emp_id, e.name, d.dept_name
FROM Employees e
FULL OUTER JOIN Departments d
ON e.dept_id = d.dept_id;
```

YE MYSQL ME NAHI CHALTA MYSQL ME FULL OUTER JOIN JESA KUCH NAHI HOTA YE POSTGRE ME CHALEGA

OR

```
SELECT e.emp_id, e.name, d.dept_name
FROM Employees e
LEFT JOIN Departments d
ON e.dept_id = d.dept_id

UNION

SELECT e.emp_id, e.name, d.dept_name
FROM Employees e
RIGHT JOIN Departments d
ON e.dept_id = d.dept_id;
```

emp_id	name	dept_name
1	Ram	HR
2	Shyam	IT
3	Mohan	NULL
4	Sita	NULL
NULL	NULL	Finance

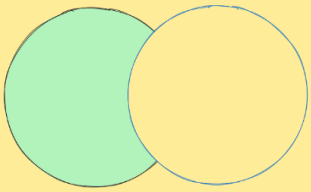
5. CROSS JOIN

-- TABLE 1 ME 4 ROWS HAI OR TABLE 2 ME 3 ROWS TO OUTPUT ME 4 * 3 12 ROWS AAYEGI

-- SYNTEX :-
```
SELECT e.name, d.dept_name
FROM Employees e
CROSS JOIN Departments d;
```

name	dept_name
Ram	Finance
Ram	IT
Ram	HR
Shyam	Finance
Shyam	IT
Shyam	HR
Mohan	Finance
Mohan	IT
Mohan	HR
Sita	Finance
Sita	IT
Sita	HR

6. LEFT ANTI JOIN



SIRF GREEN ME JITNI VALUE HAI VHI AAYEEGI

-- LEFT JOIN ME TABLE 1 JO VALUE HAI OR RIGHT ME NAHI HAI WO

-- SYNTEX :-
```
SELECT e.*
FROM Employees e
LEFT JOIN Departments d
ON e.dept_id = d.dept_id
WHERE d.dept_id IS NULL;
```

emp_id	name	dept_id
3	Mohan	NULL
4	Sita	104

7. LEFT SEMI JOIN

-- SIRF DONO TABLE ME EXIST KRTI VALUE HI DIKHEGI

-- SYNTEX :-
```
SELECT e.*
FROM Employees e
WHERE EXISTS (
  SELECT 1
  FROM Departments d
  WHERE e.dept_id = d.dept_id
);
```

SIRF EXIST KARTI VALUE HI DIKHEGI

emp_id	name	dept_id
1	Ram	101
2	Shyam	102

8. SELF JOIN

TABLE :- EMPLOYEES

emp_id	name	manager_id
1	Ram	NULL
2	Shyam	1
3	Mohan	1
4	Sita	2
5	Gita	2
NULL	NULL	NULL

-- SELF JOIN ME EK HI TABLE KO DO DIFFERNT ALIAS DEEE KAR UNHE KHUD SE KHUD KO HI JOIN KRTE HAI

-- JESE IS E MANAGER_ID HAI WO EMP_ID KE SATH JOIN KRTE HAI JESE KI EMP_ID 1 TO 5 HAI OR MANAGER_ID ME JO 1 AND 2 HAI YE WO DIKHATE HAI KI JO 1 EMP_ID VALA BANDA HAI RAM HO SHYAM AND MOHAN KA MANAGER HAI OR 2 UMBER WLAA SHAYM SITA OR GITA KA MANAGER HAI

-- SYNTEX :-
```
SELECT e.name AS employee, m.name AS manager
FROM Employees e
LEFT JOIN Employees m
ON e.manager_id = m.emp_id;
```

employee	manager
Ram	NULL
Shyam	Ram
Mohan	Ram
Sita	Shyam
Gita	Shyam